

ZHARYK, KazakhstanWMO: **354970**Lat: **48.850N**Lon: **72.867E**Elev: **656**StdP: **93.69**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.9	-26.5	-33.0	0.2	-29.5	-29.7	0.3	-26.0	11.8	-10.5	9.2	-7.1	0.6	70	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	31.9	17.3	30.0	16.6	28.3	16.1	18.8	27.9	17.9	27.0	17.2	25.9	1.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.4	21.6	14.8	11.4	20.7	13.7	10.6	20.6	56.3	28.4	53.3	27.1	50.8	26.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.2	5.2	-33.2	34.6	3.6	2.1	-35.8	36.2	-37.9	37.4	-39.9	38.6	-42.5	40.2		
(5)			-33.3	20.8	3.6	2.8	-35.9	22.7	-38.0	24.3	-40.0	25.9	-42.6	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-13.2	-12.0	-4.3	6.9	13.4	18.8	19.8	18.7	12.6	4.9	-3.8	-10.6		
	DBStd	13.27	6.22	6.42	6.51	5.77	4.44	4.08	3.71	4.20	5.07	5.50	6.78	6.53		
	HDD10.0	3180	718	617	444	124	19	1	0	1	31	173	414	639		
	HDD18.3	5309	976	850	702	343	160	44	27	48	182	416	664	897		
	CDD10.0	1119	0	0	1	31	125	266	304	269	108	16	0	0		
	CDD18.3	207	0	0	0	1	7	58	73	58	10	0	0	0		
	CDH23.3	2421	0	0	0	16	116	654	792	690	153	2	0	0		
	CDH26.7	777	0	0	0	2	19	196	281	238	41	0	0	0		
(14) Wind	WSAvg	1.9	2.0	2.2	2.3	2.0	2.0	1.7	1.6	1.6	1.5	1.7	1.7	2.0		
(15) Precipitation	PrecAvg	345	29	23	28	24	34	32	48	18	14	27	34	33		
	PrecMax	467	59	43	67	62	88	112	140	40	55	71	71	101		
	PrecMin	233	4	7	5	0	9	1	12	0	0	5	9	11		
	PrecStd	66	14	10	15	16	22	24	29	12	13	17	17	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	2.2	14.0	25.2	29.0	32.8	34.8	33.8	30.7	22.7	13.2	2.2		
		MCWB	0.0	0.2	7.0	12.5	16.1	18.6	18.3	17.2	15.1	10.9	7.3	0.3		
	2%	DB	-0.8	0.3	9.1	21.7	26.2	30.7	32.0	31.5	27.3	19.6	9.8	0.7		
		MCWB	-2.0	-1.5	5.0	11.2	14.5	17.1	17.7	16.8	13.6	9.2	5.8	-0.9		
	5%	DB	-2.8	-1.5	6.0	19.0	24.3	29.0	29.7	29.4	24.6	16.3	6.9	-0.8		
		MCWB	-4.1	-3.0	3.2	9.9	13.4	16.4	17.1	16.2	12.9	7.7	3.7	-2.1		
	10%	DB	-4.4	-3.1	3.8	16.4	22.4	27.2	27.7	27.3	22.0	13.4	4.4	-2.3		
		MCWB	-5.6	-4.4	1.7	8.7	12.6	16.0	16.5	15.7	11.7	6.6	1.7	-3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.7	7.6	13.5	17.8	23.0	20.8	19.0	16.6	11.9	8.9	1.0		
		MCDB	1.2	1.8	12.4	22.8	26.1	26.4	27.5	28.8	27.5	20.0	11.2	1.9		
	2%	WB	-1.9	-1.3	5.2	11.9	15.5	18.6	18.9	17.9	14.7	10.0	6.0	-0.6		
		MCDB	-0.7	-0.1	8.5	20.5	23.5	27.2	28.4	28.5	25.4	17.8	9.6	0.3		
	5%	WB	-3.9	-2.9	3.4	10.5	14.3	17.5	18.1	17.1	13.3	8.6	3.9	-2.0		
		MCDB	-2.9	-1.6	5.9	17.9	22.1	26.4	26.9	27.1	23.0	14.8	6.7	-0.8		
	10%	WB	-5.5	-4.4	1.8	9.2	13.2	16.6	17.4	16.3	12.2	7.2	2.0	-3.7		
		MCDB	-4.5	-3.1	3.7	15.5	20.9	25.3	25.6	25.7	20.9	12.4	4.3	-2.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.4	8.5	11.5	12.8	12.6	12.1	13.1	13.1	10.6	8.1	8.1		
		MCDBR	6.7	8.2	9.5	15.0	15.3	14.6	14.8	15.5	16.2	15.2	10.5	7.8		
		MCWBR	6.2	7.4	6.7	7.9	7.2	6.1	5.6	5.7	7.0	8.0	7.3	6.6		
		MCDBR	6.5	7.7	8.9	14.1	13.4	13.0	13.0	14.1	15.2	13.3	9.6	7.3		
	5% WB	MCWBR	6.1	7.3	6.6	7.6	7.0	6.5	6.0	6.0	7.2	7.5	7.2	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.317	0.367	0.401	0.405	0.405	0.424	0.386	0.341	0.332	0.318	0.289		
	taud		2.090	2.026	2.016	2.139	2.187	2.254	2.208	2.295	2.378	2.323	2.175	2.094		
	Ebn at Noon		744	805	826	843	855	857	832	850	859	796	698	678		
	Edh at Noon		86	118	142	140	139	132	136	119	99	87	80	75		
(44) All-Sky Solar Radiation	RadAvg		1.70	2.90	4.14	5.06	6.38	6.71	6.19	5.75	4.44	2.53	1.37	1.23		
	RadStd		0.19	0.12	0.32	0.43	0.50	0.46	0.46	0.39	0.29	0.31	0.12	0.13		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-104	-37

Nomenclature: See separate page